

Apply filters to SQL queries

Project description

In this activity, I used SQL filters to investigate login activity and retrieve employee information for security updates. By querying the log_in_attempts and employees tables, I applied conditions with AND, OR, NOT, and LIKE to return the exact data needed for each scenario.

Retrieve after hours failed login attempts

To investigate a security incident that occurred after business hours, I retrieved all failed login attempts that happened after 18:00. I used the following SQL query:

```
SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = FALSE;
```

This query filters results to show login attempts that occurred after 18:00 and were unsuccessful, helping identify possible suspicious activity.

Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09, so I retrieved login attempts from that date and the day before. I used the following query:

```
SELECT * FROM log_in_attempts WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

The OR operator returns login attempts from either date, helping with the investigation timeline.

Retrieve login attempts outside of Mexico

To investigate suspicious login activity occurring outside of Mexico, I filtered for attempts where the country was not Mexico. Because Mexico appears as both MEX and MEXICO, I used LIKE with a wildcard:

```
SELECT * FROM log_in_attempts WHERE NOT country LIKE 'MEX%';
```

This filters out all Mexico login attempts and returns only attempts from other countries.

Retrieve employees in Marketing

My team needed to update employee machines for employees who work in the Marketing department in the East building. I used this SQL query:

```
SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';
```

The AND operator ensures results include only Marketing employees located in any office beginning with 'East'.

Retrieve employees in Finance or Sales

Next, I retrieved employees in the Finance or Sales departments for additional updates. I used the following query:

```
SELECT * FROM employees WHERE department = 'Finance' OR department = 'Sales';
```

The OR operator returns employees from either department, ensuring no one is missed.

Retrieve all employees not in IT

Finally, I identified all employees who are not in the Information Technology department so they could receive security updates. I used the following query:

```
SELECT * FROM employees WHERE NOT department = 'Information Technology';
```

The NOT operator excludes employees in IT, returning only those who still need updates.

Summary

In this activity, I created SQL queries using filters to retrieve precise information needed for security investigations and system updates. I applied AND, OR, NOT, and LIKE operators to filter login attempts and employee data across two tables. These filtering skills are essential for analyzing data efficiently and supporting cybersecurity efforts.